KARAN JAGANI

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Profile

As a Computer Science student with strong programming and problem-solving skills, I am passionate about exploring software that powers next-generation computing — from embedded systems to AI-driven applications. I enjoy learning new technologies, collaborating with teams, and contributing creative solutions to real-world problems.

Educational Qualification

Anglia Ruskin University Sep 2023 - Dec 2026
BEng. (HONS) Computer Science
Modules: Software Engineering, Database design & Implementation, Software Principles, Data Structures & Algorithm Analysis, Operating systems

Technical Skills

Programming Languages: Java, C,C++, C# , Python SQL , JavaScript, HTML5, CSS
Tools & Frameworks: MATLAB, Git/GitHub, Visual Studio, VS Code, MySQL
Concept & Core Competencies: OOP , Data structures & algorithms , Operating systems
AI & Machine Learning: Basic knowledge of Machine Learning models and Neural Networks; familiar with Python libraries (NumPy, Pandas, TensorFlow, PyTorch)
Learning Interests: Embedded Systems, Machine Learning, DevOps, Cloud Platforms

Projects & Academic Experience

Real-Time Disaster Response Optimisation Using Edge-AI on Autonomous Drones

Computer Research Methodologies Project
Jan. 2025 – Mar. 2025 Anglia Ruskin University, Cambridge

- Proposed a disaster response system using **Edge-AI** for real-time drone navigation and decision-making.
- Developed a project plan with market analysis, **WBS**, **Gantt chart**, and **risk assessment**.
- Addressed research ethics and designed a **methodology** for AI performance evaluation and data analysis.

Team-Based C# Application Development with SQL Integration

Software Engineering Module Project
Sep. 2024 – Nov. 2024 Anglia Ruskin University, Cambridge

- Collaborated on a team of four to develop a fully functional C# application using Visual Studio, implementing SQL for database operations.
- Applied **OOP** principles (**encapsulation, inheritance, polymorphism**) to ensure modular, maintainable code.
- Produced software design documentation: **ER diagrams**, **UML** models, and user interface wireframes.
- Completed full project cycle: requirement analysis, implementation, UI design, testing, and maintenance planning, following structured software engineering principles

SQL Database Design and Implementation for Together Culture

Database Systems Module Project

Sep. 2024 – Nov. 2024 Anglia Ruskin University, Cambridge

- **Collaborated** on a team to analyse business requirements and design an Extended Entity Relationship Diagram (**EERD**) and schema for a live community-led company case study.
- Individually developed and implemented a relational MySQL database, inserting sample data and creating key structures with CREATE TABLE statements.
- Designed and implemented a relational **MySQL** database for a real-world business case.

Personal Projects

AI-Based Skin Image Analysis for Neonatal Jaundice Detection

Independent Research Project – Ongoing (2025)

- Developing a vision-based system to estimate skin colour for neonatal jaundice detection using **MATLAB** and **machine learning**.
- Built an image-processing pipeline to remove pores, hairs, and blemishes before colour classification.
- Prototyped algorithms in MATLAB and **deployed the system on a Raspberry Pi** using **OpenCV** for real-time image capture and analysis.

Work Experience

Customer Assistant (Sainsbury's, Cambridge, UK)

Oct 2023 – Present

- Operate tills efficiently, assisting customers with purchases and providing excellent service.
- Work both independently and within a team, using technology and initiative to deliver a smooth shopping experience.

Student Ambassador (Anglia Ruskin University)

Oct 2023 – Present

- Represent ARU at Applicant Days and Open Days, supporting event logistics, visitor engagement, and campus tours.
- Provide information about university life, accommodation, and academic programs, ensuring a welcoming and professional experience.

Visa Administrator (MyCalculus Visa Consultant services, Rajkot, India)

Jan 2023 – July 2023

- **Processed** visa **applications**, **ensuring** compliance with immigration **regulations** and accurate documentation.
- Verified client records and maintained data accuracy across internal systems.

Languages

- **English**: Fluent, **Gujarati**: Native, **Hindi**: Conversational